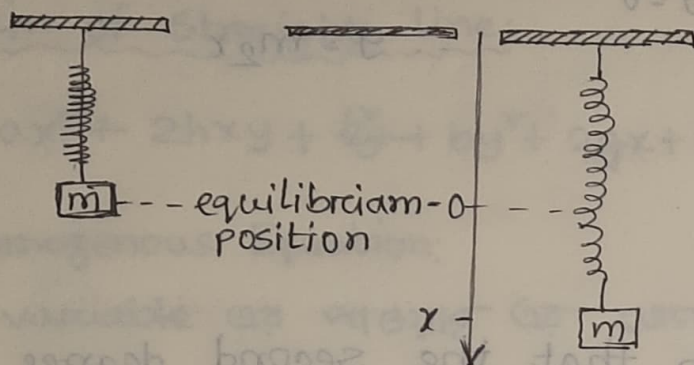


MATH 1213: ODE (Not-class)

Motion of Spring:

→ ODE. [1.0]



according to hooks law:

restoring force $\propto -kx$

If we ignore any external restoring force due to air resistance or friction,

$$\Rightarrow m \frac{d^2x}{dt^2} = -kx \quad [\text{by Newton's 2nd law}]$$

$$\Rightarrow \frac{d^2x}{dt^2} = -\frac{k}{m}x$$

Second-order differential equation

$$\Rightarrow \frac{d^2x}{dt^2} + \omega^2 x = 0$$

এখানে যেহেতু সমাধান করে $x = x(t)$ হবে বসায় যায়।

General Differential Equation:

Q. Find a solution of the differential equation

$y' = \frac{1}{2}(y^2 - 1)$ that satisfy initial condition

$$y(0) = 2.$$

$$y = \frac{1+ce^t}{1-ce^t}$$

$$\text{by } t=0 \quad y(t)=2 \Rightarrow$$

$$2-2c = 1+c$$

$$\therefore c = \frac{1}{3}$$

$$y = \frac{1+\frac{1}{3}e^t}{1-\frac{1}{3}e^t}$$

$$\therefore y = \frac{3+e^t}{3-e^t}$$

Linear Equation:

A first order linear differential equation:

$$\frac{dy}{dx} + P(x)y = Q(x)$$

* Classification of differential equation:

order: max order of derivative in an equation

degree: power or degree of that max order derivative term. (max)

eg: $\left(\frac{dy}{dx}\right)^{100} + 5\left(\frac{d^2y}{dx^2}\right)^4 + \left(\frac{d^3y}{dx^3}\right)^7 + 7y + \sin x = 0$

= order: 3 degree: 2

$$\left(\frac{d^2y}{dx^2}\right)^2 + \frac{d^3y}{dx^3} + \frac{d^5y}{dx^5} + \left(\frac{d^5y}{dx^5}\right)^5 = 0$$

order: 5 degree: 5

Non-Linear: exist product term of y & derivative of y .

eg. $y \times \frac{dy}{dx}$, $\left(\frac{dy}{dx}\right)^3 \rightarrow$ linear π

Linear: nth order linear D.E:

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1(x) \frac{dy}{dx} + a_0(x)y = F(x)$$

$$\sum_{i=0}^n a_i(x) \frac{d^i y}{dx^i} = F(x) \rightarrow \text{general structure of } n^{\text{th}} \text{ order D.E}$$

linear equation
linear or homogeneous
arbitrary constant
independent solution

Linear differential equation of order n :

$$a_0(x)y + a_1(x)\frac{dy}{dx} + a_2(x)\frac{d^2y}{dx^2} + \dots + a_n(x)\frac{d^ny}{dx^n} = F(x)$$

Homogenous: if $F(x) = 0$

Non-homogenous: if $F(x) \neq 0$

● Separable equation:

$$\frac{dy}{dx} = xy$$

$$\Rightarrow \int \frac{1}{y} dy = \int x dx$$

$$\Rightarrow \ln y = \frac{x^2}{2} + c$$

$$\therefore y = e^{\frac{x^2}{2} + c}$$

constant c at the end

$$\ln y = \frac{x^2}{2} + \ln c$$

$$\Rightarrow \ln\left(\frac{y}{c}\right) = \frac{x^2}{2}$$

$$\Rightarrow \frac{y}{c} = e^{x^2/2}$$

$$\therefore y = ce^{x^2/2}$$

Reducible to separable:

$$\frac{dy}{dx} = \text{a homogeneous function of } x, y$$

Homogeneous: if $f(tx, ty) = t^n f(x, y)$

$$1. \quad \frac{dy}{dx} = \frac{7x+8y}{9x-10y}$$

Let $y = vx$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{7x+8vx}{9x-10vx}$$

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{7+8v}{9-10v} - v = \frac{7+8v-9v+10v^2}{9-10v}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{10v^2 - v + 7}{9-10v}$$

$$\Rightarrow \dots = \ln x$$

$$\Rightarrow \int \frac{9-10v}{10v^2-v+7} dv = \int \frac{dx}{x}$$

$v \rightarrow y/x$ complete

Integrating factor:

1st order linear D.E:

inexact

$$x^r \frac{dy}{dx} + 2y = x^r$$

$$\rightarrow \frac{dy}{dx} + P(x)y = Q(x)$$

$$\Rightarrow x^r \frac{dy}{dx} + 2xy = x^3$$

Integrating Factor (IF):

$$\Rightarrow \frac{d}{dx} (x^r y) = x^3$$

$$= e^{\int P(x) dx}$$

IF first multiply

$$\Rightarrow x^r y = \int x^3 dx$$

$$P(x) = \frac{2}{x} \quad IF = e^{\int \frac{2}{x} dx}$$

$$\Rightarrow x^r y = \frac{x^4}{4} + C$$

$$IF = e^{2 \int \frac{1}{x} dx} = e^{2 \ln x} = e^{(\ln x)^2}$$

$$\therefore y = \frac{x^r}{4} + \frac{C}{x^r}$$

$$(*) (IF) \frac{dy}{dx} + (IF)P(x)y = (IF)Q(x)$$

exact

$$2. \frac{dy}{dx} + x^3 y = x$$

$$IF = e^{\int x^3 dx} = e^{\frac{x^4}{4}}$$

$$\Rightarrow e^{\frac{x^4}{4}} \frac{dy}{dx} + e^{\frac{x^4}{4}} x^3 y = x e^{\frac{x^4}{4}}$$

$$\Rightarrow \frac{d}{dx} [y e^{\frac{x^4}{4}}] = x e^{\frac{x^4}{4}}$$

$$\Rightarrow y e^{\frac{x^4}{4}} = \int x e^{\frac{x^4}{4}} dx$$

$$\Rightarrow y$$

● Exact differential equation:

$$M(x, y) dx + N(x, y) dy = 0 \quad \downarrow$$

example:

$$y dx + x dy = 0$$

$$\Rightarrow d(xy) = 0$$

$$\Rightarrow \int d(xy) = 0$$

$$\Rightarrow xy = c \quad \text{(Ans.)}$$

Next, $f(x)$ continuous

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y}$$

$$\Rightarrow M(x, y) dx + N(x, y) dy = 0$$

$$\Rightarrow \frac{\partial f}{\partial x} dx + \frac{\partial f}{\partial y} dy = 0$$

$$\Rightarrow \frac{d}{dx} [f(x, y)] = 0$$

$$M(x, y) = \frac{\partial f}{\partial x} \quad N(x, y) = \frac{\partial f}{\partial y}$$

$$\Rightarrow \frac{\partial M}{\partial y} = \frac{\partial^2 f}{\partial y \partial x} \Rightarrow \frac{\partial N}{\partial x} = \frac{\partial^2 f}{\partial x \partial y}$$

$$\text{so, } \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \Rightarrow \boxed{M_y = N_x}$$

The differential equation

$$M(x, y) dx + N(x, y) dy = 0$$

is exact if $M_y = N_x$

eg 1: solve $(2xy - 3x^2) dx + (2y + x^2 + 1) dy = 0$; $y(0) = 3$

$$M(x, y) = 2xy - 3x^2 \quad M_y = 2x$$

$$N(x, y) = 2y + x^2 + 1 \quad N_x = 2x$$

as $M_y = N_x$ exact equation.

$$\text{Now, } M = \frac{\partial f}{\partial x}$$

$$N = \frac{\partial f}{\partial y}$$

$$\Rightarrow 2xy - 3x^2 = \frac{\partial f}{\partial x} \quad \text{--- (i)}$$

$$\Rightarrow 2y + x^2 + 1 = \frac{\partial f}{\partial y} \quad \text{--- (ii)}$$

$$\Rightarrow f(x, y) = \int (2xy - 3x^2) dx$$

$$\Rightarrow f(x, y) = \int (2y + x^2 + 1) dy = y^2$$

$$\Rightarrow f(x, y) = x^2 y - 3x^3 + 2y \quad \text{--- (3)}$$

From ① $\Rightarrow \frac{\partial F}{\partial y} = 2y + x + 1$

$\Rightarrow \frac{\partial f}{\partial y} [x^2y - 3x^3 + g(y)] = 2y + x + 1$

$\Rightarrow x^2 \cdot 1 - 0 + \frac{d}{dy} g(y) = 2y + x + 1$

$\Rightarrow \frac{d}{dy} g(y) = 2y + 1$ এখানে যদি x এর টার্ম
আমরা তাকে ব্রুন্ডে করে
কোথায় ব্রুন্ডে হয়েছে

$\Rightarrow g(y) = \int (2y + 1) dy$

$\Rightarrow g(y) = y^2 + y + c$

$g(y) \dots \dots \text{③} \Rightarrow f(x, y) = x^2y - 3x^3 + y^2 + y$

$\Rightarrow x^2y - 3x^3 + y^2 + y = c$

as, $y(0) = -3 : 0 - 3 \cdot 0^3 + 3^2 + (-3) = c$

$\therefore c = 6$

$\therefore$ The solution: $x^2y - 3x^3 + y^2 + y = 6$ Ans.

Now, $Mdx + Ndy = 0$ is not exact:

then do the following:

Integrating Factor $\left\{ \begin{array}{l} e^{\int \left(\frac{M_y - N_x}{N} \right) dx} ; \frac{M_y - N_x}{N} = f(x) \\ e^{\int \left(\frac{N_x - M_y}{M} \right) dy} ; \frac{N_x - M_y}{M} = f(y) \end{array} \right.$

$\rightarrow (IF \times M) dx + (IF \times N) dy = 0$

(exact now)

● Higher order differential equation:

Q1. $\frac{d^2y}{dx^2} - 5 \frac{dy}{dx} + 6y = 0$

Let $y = e^{mx}$; $m^2 e^{mx} - 5m e^{mx} + 6e^{mx} = 0$

$\Rightarrow e^{mx} [m^2 - 5m + 6] = 0$

as, $m^2 - 5m + 6 = 0$

$\Rightarrow (m-3)(m-2) = 0$

$\therefore m = 3, 2$

$y = ce^{2x}$, $y = ce^{3x}$ both works.

* If $y = e^{2x}$ is a solution then so is $y = ce^{2x}$ for any $c \in \mathbb{R}$.

* If $y = f_1(x)$ $y = f_2(x)$ are solution then so is $y = f_1(x) + f_2(x)$

● If $y = f_1(x)$ and $y = f_2(x)$ are two solutions then the general solution is

$y = c_1 f_1(x) + c_2 f_2(x)$ Linear combination

$y = c_1 e^{2x} + c_2 e^{3x}$ Ans.

Q2. $\frac{d^3y}{dx^3} - 6 \frac{d^2y}{dx^2} + 11 \frac{dy}{dx} - 6y = 0$

A.E: $m^3 - 6m^2 + 11m - 6 = 0$

$\Rightarrow m^2(m-1) - 5m(m-1) + 6(m-1) = 0$

$\Rightarrow (m-1)(m-2)(m-3) = 0$

$\therefore m = 1, 2, 3$

∴ general solution:

$$y = c_1 e^x + c_2 e^{2x} + c_3 e^{3x} \quad (\text{Ans.})$$

case 1: A.E Root real and distinct (↑)

case 2: A.E Root real & equal.

Q3: $\frac{d^2 y}{dx^2} - 6 \frac{dy}{dx} + 9y = 0$

A.E: $m^2 - 6m + 9 = 0$

$$\Rightarrow (m-3)^2 = 0$$

$$\therefore m = 3, 3$$

gen soln: $y = c_1 e^{3x} + c_2 x e^{3x}$

(method of variation of parameters: $x f(x)$)

if $m = 2, 2, 2, 2, 3, 4$

gen. soln: $y = c_1 e^{3x} + c_2 e^{4x} + c_3 e^{2x} + c_4 x e^{2x} + c_5 x^2 e^{2x} + c_6 x^3 e^{2x}$

case 3: A.E roots are complex

Q4: $\frac{d^2 y}{dx^2} + \frac{dy}{dx} + y = 0$

$$m^2 + m + 1 = 0 \Rightarrow m = \frac{-1 \pm \sqrt{3}i}{2}$$

Gen. soln: $m = a + ib$

$$y = e^{ax} [c_1 e^{i(bx)} + c_2 e^{-i(bx)}]$$

$$= e^{ax} [c_1 (\cos bx + i \sin bx) + c_2 (\cos bx - i \sin bx)]$$

Q34.

$$\frac{d^2 y}{dx^2} + \frac{dy}{dx} + y = 0$$

$$m^2 + m + 1 = 0$$

$$\therefore m = \frac{-1 \pm \sqrt{3}i}{2}$$

$$m = a + ib$$

gen soln: $y = c_1 e^{(a+ib)x} + c_2 e^{(a-ib)x}$

$$y = c_1 e^{ax} \cdot e^{ibx} + c_2 e^{ax} \cdot e^{-ibx}$$

$$= e^{ax} [c_1 e^{i(bx)} + c_2 e^{-i(bx)}]$$

$$y = e^{ax} [(c_1' + c_2') \cos bx + (ic_1' - ic_2') \sin bx]$$

$$= e^{ax} [c_1 \cos bx + c_2 \sin bx]$$

$$m = a + ib$$

$$\text{From : } m = \frac{-1 \pm \sqrt{3}i}{2} = (-\frac{1}{2}) \pm (\frac{\sqrt{3}}{2})i$$

$$\therefore \text{ gen soln: } y = e^{\frac{1}{2}x} [c_1 \cos \frac{\sqrt{3}}{2}x + c_2 \sin \frac{\sqrt{3}}{2}x]$$

$$\# m = 2, 3, 3, 4 \pm 5i, 4 \pm 5i, 6 \pm 7i$$

$$\text{gen. soln: } y = c_1 e^{2x} + c_2 e^{3x} + c_3 x e^{3x} +$$

$$e^{4x} [c_4 \cos 5x + c_5 \sin 5x] +$$

$$x e^{4x} [c_6 \cos 5x + c_7 \sin 5x] +$$

$$e^{6x} [c_8 \cos 7x + c_9 \sin 7x]$$

Non-Homogenous Linear differential equation with constant co-efficients:

$$\frac{d^2y}{dx^2} + (-7) \frac{dy}{dx} + 10y = 2x + 3 \rightarrow \text{non-homogenous}$$

Let the soln be $y = y_p(x) = y_p$

$$\hookrightarrow \frac{d^2y}{dx^2} - 7 \frac{dy}{dx} + 10y = 0 \quad \text{A.E: } m^2 - 7m + 10 = 0$$

$$\hookrightarrow y = y_c(x) = y_c$$

$$\Rightarrow m = 5, 2$$

$$\Rightarrow y_c(x) = c_1 e^{5x} + c_2 e^{2x}$$

Gen. soln:

$$y = y_c(x) + y_p(x)$$

$\hookrightarrow$ complementary function

Particular function / integral

$$\frac{d^2y}{dx^2} - 7 \frac{dy}{dx} + 10y = 2x+3$$

$$y = y_p(x) \rightarrow \frac{d^2y_p}{dx^2} - 7 \frac{dy_p}{dx} + 10y_p = 2x+3 \quad \text{--- ①}$$

$$\frac{d^2y}{dx^2} - 7 \frac{dy}{dx} + 10y = 0$$

$$y = y_c(x) \rightarrow \frac{d^2y_c}{dx^2} - 7 \frac{dy_c}{dx} + 10y_c = 0 \quad \text{--- ②}$$

$$\text{L.H.S: } \frac{d^2(y_c+y_p)}{dx^2} - 7 \frac{d(y_c+y_p)}{dx} + 10(y_c+y_p) =$$

$$\Rightarrow \frac{d^2y_c}{dx^2} - 7 \frac{dy_c}{dx} + 10y_c + \frac{d^2y_p}{dx^2} - 7 \frac{dy_p}{dx} + 10y_p$$

$$= 2x+3+0$$

$$= y$$

$$\therefore y = y_c(x) + y_p(x)$$

Example 1:

$$\text{Solve } \frac{d^2y}{dx^2} - 7 \frac{dy}{dx} + 10y = 0 + 2x+3$$

Finding $y_c(x)$ →

$$\text{A.E: } m^2 - 7m + 10 = 0$$

$$m = 2, 5$$

$$y_c = c_1 e^{2x} + c_2 e^{5x}$$

$$y_p = \frac{1}{5}x + \frac{11}{25}$$

∴ gen soln:

$$y = c_1 e^{2x} + c_2 e^{5x} + \frac{1}{5}x + \frac{11}{25}$$

Finding y_p :

Let the solution be

$$y_p = ax+b = Ax+B$$

$$\Rightarrow \frac{d^2y_p}{dx^2} - 7 \frac{dy_p}{dx} + 10y_p = 2x+3$$

$$\Rightarrow 0 - 7A + 10Ax + 10B = 2x+3$$

$$\Rightarrow 10Ax + 10B - 7A = 2x+3$$

$$\hookrightarrow 10A = 2 \therefore A = \frac{1}{5}$$

$$10B - 7A = 3 \Rightarrow 10B - \frac{7}{5} = 3$$

$$\therefore B = \frac{38}{10} = \frac{11}{25}$$

undetermined coefficients:

R.H.S = Algebra, ^{sin/cos} trig, expon, combination $\rightarrow$ (x) qk = k
 $\rightarrow$ R.S. 17
সামঞ্জস্য

$$\text{R.H.S: } x^3 \rightarrow ax^r + bx^r + cx + d$$

$$\sin x \rightarrow a \sin x + b \cos x$$

$$e^{5x} \rightarrow A e^{5x}$$

uc set:

$$\textcircled{*} x^3 \rightarrow \text{uc set} = \{x^3, x^r, x, 1\}$$

$$y_p = ax^3 + bx^r + cx + d$$

$$\textcircled{*} 2\sin x + \cos x \rightarrow \text{uc set} = \{\sin x, \cos x\}$$

$$y_p = a \sin x + b \cos x$$

$$\textcircled{*} e^{2x} \rightarrow \text{uc set} = \{e^{2x}\} \quad [\text{ত্রিমাত্রিক ক্ষেত্রে ২টি ভাৱ}]$$

$$y_p = A e^{2x}$$

$$y_p = x A e^{2x}$$

$$\textcircled{*} x \sin x \rightarrow \text{uc set} = \{x, 1\} \times \{\sin x, \cos x\}$$

$$= \{x \sin x, x \cos x, \sin x, \cos x\}$$

$$y_p = A x \sin x + B x \cos x + c \sin x + d \cos x$$

$$\textcircled{*} x e^{5x} \cos 2x \rightarrow \text{uc set} = \{x^r, x, 1\} \times \{e^{5x}\} \times \{\sin^2 2x, \cos 2x\}$$

$$= \{x^r e^{5x} \sin x, x^r e^{5x} \cos 2x, x e^{5x} \sin 2x, x e^{5x} \cos 2x, e^{5x} \sin 2x, e^{5x} \cos 2x\}$$

$$\therefore y_p = A x^r e^{5x} \sin x + B x^r e^{5x} \cos 2x + c e^{5x} \sin 2x + d e^{5x} x \cos 2x + \dots + F e^{5x} \cos 2x$$

$$\textcircled{*} \cos x = \{\sin x, \cos x\}$$

[N.B.]

$$y_c = c_1 e^{2x} + c_2 e^{5x}$$

$$\text{R.H.S} = e^{2x} \rightarrow y_p = A x e^{2x}$$

$$= x e^{2x} \rightarrow y_p = A x^2 e^{2x}$$

$$= x^2 e^{2x} \rightarrow y_p = A x^2 e^{2x} + B x e^{2x}$$

$$y_c = c_1 e^{2x} + c_2 x e^{5x} + c_3 e^{5x}$$

$$\text{R.H.S: } e^{5x} \rightarrow y_p = A x^2 e^{5x}$$

$$x e^{5x} \rightarrow y_p = A x^2 e^{5x}$$

$$x^2 e^{5x} \rightarrow y_p = A x^2 e^{5x}$$

$$x^3 e^{5x} \rightarrow y_p = A x^3 e^{5x} + B x^2 e^{5x}$$

● Application of differential equation:

1 Hook's Law:

$$F \propto -x$$

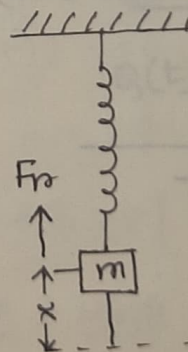
$$\Rightarrow F = -kx$$

$$\Rightarrow m \frac{d^2 x}{dt^2} = -kx \quad \frac{k}{m} = \omega^2$$

$$\Rightarrow \frac{d^2 x}{dt^2} = -\frac{k}{m} x$$

$$\Rightarrow \frac{d^2 x}{dt^2} + \omega^2 x = 0$$

$$\therefore \frac{d^2 x}{dt^2} + \omega^2 x = 0$$



$$x(0) = A$$

$$x'(0) = \Delta$$

Linear
Homogeneous
constant co-eff.

$$\frac{d^2 x}{dt^2} + \omega^2 x = 0$$

A.E: $\lambda^2 + \omega^2 = 0$

$$\Rightarrow \lambda^2 = -\omega^2$$

$$\Rightarrow \lambda = \pm i\omega$$

$$x = e^{at} (c_1 \sin \omega t + c_2 \cos \omega t)$$

$$x = c_1 \sin(\omega t) + c_2 \cos(\omega t)$$

$$t=0 : x(0) = A \quad x'(0) = 0$$

$$\textcircled{1} c_1 \sin(0) + c_2 \cos(0) = A \quad \therefore c_2 = A$$

$$\textcircled{II} \frac{d}{dt} [c_1 \sin(\omega t) + c_2 \cos(\omega t)] = 0$$

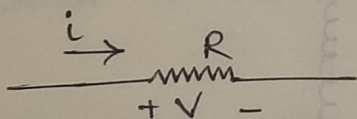
$$\Rightarrow c_1 \omega \cos(\omega t) - A \sin(\omega t) \cdot \omega = 0$$

$$\Rightarrow c_1 \omega \cos(0) - A \sin(0) \cdot \omega = 0$$

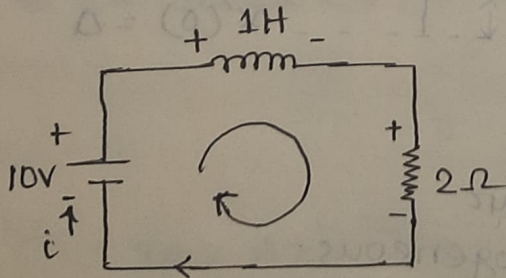
$$\therefore c_1 = 0$$

$$\therefore x(t) = A \cos(\omega t)$$

2



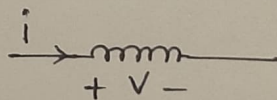
$$V = iR$$



KVL:

$$-10 + 1 \frac{di}{dt} + 2i = 0$$

$$\Rightarrow \frac{di}{dt} = -2i + 10$$



$$V \propto \frac{di}{dt}$$

$$V = L \frac{di}{dt}$$

Inductance

$$\frac{di}{dt} = 10 - 2i$$

$$\Rightarrow \int \frac{di}{i-5} = -2 \int dt$$

$$\Rightarrow \ln(i-5) = -2t + \ln A$$

$$\Rightarrow \ln\left(\frac{i-5}{A}\right) = -2t$$

$$\Rightarrow i-5 = Ae^{-2t}$$

$$\therefore i = 5 + Ae^{-2t}$$

$$i(t) = 5 + Ae^{-2t}$$

capacitor:

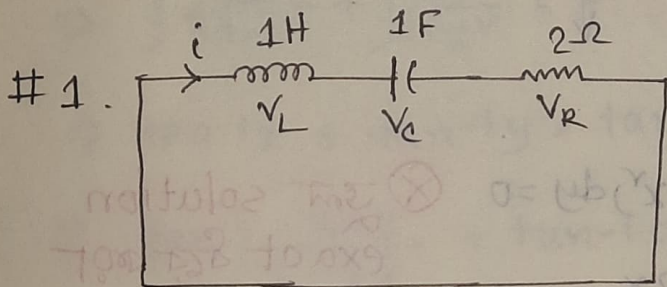
$$Q \propto V \quad \boxed{Q = CV}$$

$$i = \frac{dQ}{dt} = \frac{d}{dt}(CV)$$

$$i = C \frac{dV}{dt}$$

$$i = C \frac{dV}{dt}$$

$$i = C \frac{dV}{dt}$$



$$v_L + v_C + v_R = 0$$

$$\Rightarrow 1 \frac{di}{dt} + v_C + 2i = 0$$

$$\Rightarrow \frac{dv_C}{dt} + \frac{dv_C}{dt} + 2i \frac{di}{dt} = 0$$

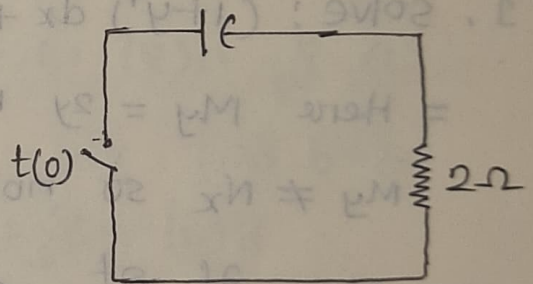
$$\Rightarrow \frac{dv_C}{dt} + 2 \frac{di}{dt} + i = 0$$

A.E: $m^2 + 2m + 1 = 0$

$$m = -1, -1$$

$$y = c_1 e^{-x} + c_2 x e^{-x}$$

#2. $Q(0) = 100 \text{ C}$



$$Q(t) = ?$$